Cs501 Quiz NO 1 2016 Complete Solved File

What functionality is performed by the instruction “lar R3, 36” of SRC?

Select correct option

It will load the register R3 with the contents of the memory location M [PC+36]

It will load the register R3 with the relative address itself (PC+36).

It will store the register R3 contents to the memory location M [PC+36]

No operation

Which type of Instruction is stored in instruction register (IR)?

Select correct option

Current instruction

Next instruction

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Previous instruction

Next and current instruction

Which of the following RTL descriptions is used for specifying the operation of an SRC instruction?

Select correct option

IR<31..27>

IR<22..26>

IR<21..17>

IR<21..0>

What is the size of the memory space that is available to FALCON-A processor?

Select correct option

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2^8 bytes

2^16 bytes

2^32 bytes

2^64 bytes

From the given subsets of the systems, to which subset does a computer belong?

Select correct option

Mechanical System

Electrical system

Optical system

Biological system

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From the given subsets of the systems, to which subset does a computer belong?

Select correct option

Mechanical System

Electrical system

Optical system

Biological system

In SRC, all the instructions are _____ bits long.

Select correct option

8

16

32

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64

In SRC, the effective address is computed at run-time by adding a constant to value of _____ register .

Select correct option

FLAGS

IR

PC

Ra

P: R3 <- R5

MAR <- IR

These two are instructions written using RTL.If these two operations is to occur simultaneously then which symbol will we use to separate them so that it becomes a correct statement with the condition that two operations occur simultaneously?

Select correct option

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Parentheses ()

Arrow <-

Colon :

Comma ,

What does the instruction “ldr R3, 58” of SRC do?

Select correct option

It will load the register R3 with the contents of the memory location M [PC+58]

It will load the register R3 with the relative address itself (PC+58).

It will store the register R3 contents to the memory location M [PC+58]

No operation

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A collection of all possible machine language commands that a computer can understand

and execute is called _____.

Select correct option

Bytecodes

Instruction Set

Binary codes

None of the given

Which field of the machine language instruction is the "type of operation" that is to be performed?

Select correct option

Op-code (or the operation code)

CPU registers

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Memory cells

I/O locations

Which one of the following registers stores a previously calculated value or a value loaded from the main memory?

Select correct option

Accumulator

Address Mask

Instruction Register

Program Counter

Which of the following bits of SRC instruction are used to hold an operand, an address index, or a branch target register?

Select correct option

The bits 16 through 0

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The bits 26 through 22

The bits 21 through 17

The bits 17 through 0

In a _____, the address of next instruction to be executed is also given with the current instruction.

Select correct option

1-address instruction

2-address instruction

3-address instruction

4-address instruction

From the given subsets of the systems, to which subset does a computer belong?

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Select correct option

Mechanical System

Electrical system

Optical system

Biological system

Which one of the following is the highest level of abstraction in digital design in which the computer architect views the system for the description of system components and their interconnections?

Select correct option

Processor-Memory-Switch level (PMS level)

Instruction Set Level

Register Transfer Level

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Control Tranfer Level

Which instruction is used to store register to memory using relative address?

Select correct option

ld instruction

ldr instruction

lar instruction

str instruction

In----------address mode, the actual data is stored in the instruction.

Select correct option

Direct

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Indirect

Immediate

Relative

How can we refer to an instruction register (IR), of 16 bits (numbered 0 to 15) using RTL?

Select correct option

IR<16..0>

IR<15..0>

IR<16..1>

IR<15..1>

A machine cycle is defined to be the time it takes to fetch _____ operands from registers, perform an Arithmetic logic Unit (ALU) operation and store the result in a register.

Select correct option

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Three

Two

One

Four

Which of the instruction is used to load register from memory using a relative address?

Select correct option

ld instruction

ldr instruction

lar instruction

str instruction

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_____ is a Miscellaneous instruction.

Select correct option

Shiftl

Store

Halt

Call

The size of data bus of MC68000 processor is _____.

Select correct option

8 bits

16 bits

20 bits

32 bits

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Which one of the following registers holds the address of the next instruction to be executed?

Select correct option

Accumulator

Address Mask

Instruction Register

Program Counter

What is the instruction length of the FALCON-E processor?

Select correct option

8 bits

16 bits

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32 bits

64 bits

Which notation do we use to name different fields of a register in RTL?

Select correct option

()

<-

+

:=

Which type of instructions enables mathematical computations?

Select correct option

Arithmetic

Waham Shah Zili

Control

Data transfer

Numeric

A collection of all possible machine language commands that a computer can understand

and execute is called _____.

Select correct option

Bytecodes

Instruction Set

Binary codes

None of the given

What functionality is performed by the instruction “str R8, 34” of SRC?

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Select correct option

It will load the register R8 with the contents of the memory location M [PC+34]

It will load the register R8 with the relative address itself (PC+34).

It will store the register R8 contents to the memory location M [PC+34]

No operation

The instruction ----------will load the register R3 with the contents of the memory location M [PC+56]

Select correct option

Add R3, 56

lar R3, 56

ldr R3, 56

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str R3, 56

Which operator is used to ‘name’ registers, or part of registers, in the Register Transfer Language?

Select correct option

:=

&

%

©

Which of the following bits of SRC instruction are used as short displacement or to specify an immediate operand?

Select correct option

c2<11..0>: = IR<11..0>;

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c2<21..0>: = IR<21..0>;

c2<16..0>: = IR<16..0>;

c2<11..0>: = IR<11..0>;

In RTL, which of the following symbols is used to store some data into a register?

Select correct option

:

<-

:=

==

In----------address mode, the actual data is stored in the instruction.

Select correct option

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Direct

Indirect

Immediate

Relative

mul is the example of a(n) _____ operation.

Select correct option

Logic

Shift

Arithmetic

Data transfer

In SRC, all the instructions are _____ bits long.

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Select correct option

8

16

32

64

“If P = 1, then load the contents of register R1 into register R2”. This statement can be written in RTL as:

Select correct option

R1 <- R2

P: R1 <- R2

P: R2 <- R1

P: R2 <- R1, P: R1 <- R2

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Execution Time = IC x CPI x MIPS

Execution Time = IC x CPI x T

Execution Time = CPI x T x MFLOPS

Execution Time = IC x T

Which type of instructions load data from memory into registers, or store data from registers into memory and transfer data between different kinds of special-purpose registers?

Select correct option

Arithmetic

Control

Data transfer

Maham Shah Zill

Floating point

What is the instruction length of the FALCON-A processor?

Select correct option

8 bits

16 bits

32 bits

64 bits

Which of the following bits of SRC instruction are used to hold an operand, an address index, or a branch target register?

Select correct option

The bits 16 through 0

The bits 26 through 22

The bits 21 through 17

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The bits 17 through 0

Which one of the following is the memory organization of EAGLE processor?

Select correct option

2^8 * 8 bits

2^16 * 8 bits

2^32 * 8 bits

2^64 * 8 bits

An “assembler” that runs on one processor and translates an assembly language program written for another processor into the machine language of the other processor is called a ----------

Select correct option

compiler

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cross assembler

debugger

linker

_____ instruction is used to load a register with an immediate data value.

Select correct option

la

lar

ld

none of the given

SPEC stands for _____.

Select correct option

Scientific Performance Execution Cooperative

Maham Shah Zill

System Performance Execution Cooperative

Scientific Performance Evaluation Cooperative

System Performance Evaluation Cooperative

Which of the following RTL descriptions is used to represent all general purpose registers of SRC?

Select correct option

R [1..32]<32..1>;

R [1..32]<31..0>;

R [0..31]<32..1>;

R [0..31]<31..0>;

Motorola MC68000 is an example of ---------microprocessor.

Select correct option

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CISC

RISC

SRC

FALCON

_____ is a register which takes input from the ALSU as a memory address to be accessed and transfers the memory contents on that location onto the memory sub-system.

Select correct option

PC

MBR

MAR

IR

Which one of the following is a bi-stable device, capable of storing one bit of Information?

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Select correct option

Decoder

Flip-flop

Multiplexer

Diplexer

The size of data bus of MC68000 processor is _____.

Select correct option

8 bits

16 bits

20 bits

32 bits

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The instruction "PUSH A" is an example of _____.

Select correct option

0-address instruction

1-address instruction

2-address instruction

3-address instruction

In register-based machines, a _____ field is required in the instruction to identify a register.

Select correct option

4-bit

6-bit

8-bit

Mahan Shah Zill

16-bit

In a _____, the address of next instruction to be executed is also given with the current instruction.

Select correct option

1-address instruction

2-address instruction

3-address instruction

4-address instruction

In Type C instruction of SRC, the maximum size of constant field is _____ bits.

Select correct option

8

16

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17

27

Which one of the following registers stores a previously calculated value or a value loaded from the main memory?

Select correct option

Accumulator

Address Mask

Instruction Register

Program Counter

Which type of instructions load data from memory into registers, or store data from registers into memory and transfer data between different kinds of special-purpose registers?

Select correct option

Maham Shah Zill

Arithmetic

Control

Data transfer

Floating point

A _____ is a computer program used to aid in detecting errors in a program.

Select correct option

Debugger

Assembler

Linker

Compiler

Maham Shah Zill

Type D instruction format of SRC includes _____ ALU registers.

Select correct option

3

4

1

2

Execution time of a program with respect to the processor is calculated as:

Select correct option

Execution Time = IC x CPI x MIPS

Execution Time = IC x CPI x T

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Execution Time = CPI x T x MFLOPS

Execution Time = IC x T

In CPU design, _____ creates or forms the interface between the data path and the control unit.

Select correct option

Busses

ALU

Control Signal

Cache

Which of the following register(s) is/are programmer invisible and is/are required to hold an operand or result value while the bus is busy transmitting some other value?

Select correct option

Instruction Register

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Memory address register

Memory Buffer Register

Registers A and C

A general purpose digital computer has _____ main components.

Select correct option

2

3

4

5

RISC stands for?

Select correct option

Maham Shah Zill

Registers internal system cache

Reduced Instruction Set Computer

Register Instruction Set Computer

Reduced Internal System Computers

In RTL, which of the following symbols is used to store some data into a register?

Select correct option

:

<-

:=

==

Maham Shah Zill

Which notation do we use to name different fields of a register in RTL?

Select correct option

()

<-

+

:=

Which of the following register(s) is/are programmer invisible and is/are required to hold an operand or result value while the bus is busy transmitting some other value?

Select correct option

Instruction Register

Memory address register

Memory Buffer Register

Maham Shah Zill

Registers A and C

In SRC, the op-code for NOP operation is _____.

Select correct option

0

2

3

6

In FALCON-A instruction format of TYPE-2 constants and variable should be in the range of

Select correct option

-132 to +131

-164 to +163

-32 to +31

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-128 to +127

Type A of SRC has which of the following instructions?

a) andi, instruction

b) No operation or nop instruction

c) lar instruction

d) ldr instruction

e) Stop operation or stop instruction

Select correct option

(a)& (b)

(b)&(c)

(a)&(e)

(b)&(e)

In a simple RISC computer, the size of each register is _____.

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Select correct option

16 bits

24 bits

32 bits

64 bits

A general purpose digital computer has _____ main components.

Select correct option

2

3

4

5

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_____ instruction is used to load a register with an immediate data value.

Select correct option

la

lar

ld

none of the given

In which of the following instructions, the data moves between a register in the processor and a memory location (or another register)?

Select correct option

Arithmetic/logic

Load/store

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Test/branch

None of the given

For the _____ type instructions, we require a register to hold the data that is to be loaded from the memory, or stored back to the memory

Select correct option

Jump

Control

load/store

Branch

SPEC stands for _____.

Select correct option

Scientific Performance Execution Cooperative

Maham Shah Zill

System Performance Execution Cooperative

Scientific Performance Evaluation Cooperative

System Performance Evaluation Cooperative

In SRC, the effective address is computed at run-time by adding a constant to value of _____ register .

Select correct option

FLAGS

IR

PC

Ra

Program counter (PC) holds the memory address of _____?

Select correct option

Maham Shah Zill

Previous instruction

Current instruction

Next instruction

Previous and current instruction

Which of the following RTL descriptions is used for specifying the operation of an SRC instruction?

Select correct option

IR<31..27>

IR<22..26>

IR<21..17>

IR<21..0>

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Which one of the followings is NOT related to the architecture of a computer?

Select correct option

Instruction set

Control signals

I/O mechanisms

Memory addressing modes

Which type of instructions enables mathematical computations?

Select correct option

Arithmetic

Control

Data transfer

Maham Shah Zill

Numeric

CISC Stands for?

Select correct option

Computer instruction set compiler

Complex instruction set computer

Complex internal system computer

Complex instruction system compiler

What does the word ‘D’ in the ‘D-flip-Flop’ stand for?

Select correct option

Data

Digital

Dynamic

Maham Shah Zill

Double

Which one of the following portions of an instruction represents the operation to be performed?

Select correct option

Address

Instruction code

Opcode

Operand

In FALCON-A processor memory word size is of

Select correct option

1 byte

4 byte

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2 byte

8 byte

The instruction ----------will load the register R3 with the contents of the memory location M [PC+56]

Select correct option

Add R3, 56

lar R3, 56

ldr R3, 56

str R3, 56

Program counter (PC) and Instruction register (IR) are normally 16-Bit registers, however in SRC, these are _____.

Select correct option

Maham Shah Zill

31-Bit

16-Bit

32-Bit

64-Bit

How much connections are required in a “point-to-point” scheme, whereas we have five (5) m-bit registers.

Select correct option

25

30

20

24

Maham Shah Zill

In SRC, the general-purpose register file includes _____ registers, each 32 bit wide.

Select correct option

16 registers R0 to R15

24 registers R0 to R23

32 registers R0 to R31

64 registers R0 to R63

SPARC uses a simple set of _____ Instruction formats.

Select correct option

64-bit

12-bit

16-bit

Maham Shah Zill

32-bit

What is the working of Processor Status Word (PSW)?

Select correct option

To hold the current status of the processor.

To hold the address of the current process

To hold the instruction that the computer is currently processing

To hold the address of the next instruction in memory that is to be executed

The instruction "PUSH A" is an example of _____.

Select correct option

0-address instruction

1-address instruction

Maham Shah Zill

2-address instruction

3-address instruction

Which type of instructions load data from memory into registers, or store data from registers into memory and transfer data between different kinds of special-purpose registers?

Select correct option

Arithmetic

Control

Data transfer

Floating point

A machine cycle is defined to be the time it takes to fetch _____ operands from registers, perform an Arithmetic logic Unit (ALU) operation and store the result in a register.

Select correct option

Maham Shah Zill

Three

Two

One

Four

Almost every commercial computer has its own particular ---------- language

Select correct option

3GL

English language

Higher level language

Assembly language

Maham Shah Zill

Which type of Instruction is stored in instruction register (IR)?

Select correct option

Current instruction

Next instruction

Previous instruction

Next and current instruction

How much connections are required in a “point-to-point” scheme, whereas we have five (5) m-bit registers.

Select correct option

25

30

20

24

Maham Shah Zill